

sorting speed in MB/s

1600
1400
1200
1000
800
600
400
200
0

2x Power9, 2.9 GHz 16C SMT4

256 GiB RAM

$\sigma = 256$ (one byte)

$n_t = 4$ (key); 4 bytes val.

$b = 512$

parallel, 4 threads

radsort, 1 thread

generic, 1 thread

bitmanip, 1 thread

swc, 1 thread

△ larger is better

1 cacheline ->

L1 cache ->

L2 cache ->

L3 cache ->

RAM

$2^5 = 32$ B

$2^{10} = 1$ KiB

$2^{15} = 32$ KiB

$2^{20} = 1$ MiB

$2^{25} = 32$ MiB

$2^{30} = 1$ GiB

$2^{35} = 32$ GiB

array length [bytes]

